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Optimal Mean Reversion Trading with Transaction Costs and Stop-Loss Exit

[Tim Leung](#), [Xin Li](#)

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Motivated by the industry practice of pairs trading, we study the optimal timing strategies for trading a mean-reverting price spread. An optimal double stopping problem is formulated to analyze the timing to start and subsequently liquidate the position subject to transaction costs. Modeling the price spread by an Ornstein–Uhlenbeck process, we apply a probabilistic methodology and rigorously derive the optimal price intervals for market entry and exit. As an extension, we incorporate a stop-loss constraint to limit the maximum loss. We show that the entry region is characterized by a bounded price interval that lies strictly above the stop-loss level. As for the exit timing, a higher stop-loss level always implies a lower optimal take-profit level. Both analytical and numerical results are provided to illustrate the dependence of

timing strategies on model parameters such as transaction cost and stop-loss level.

Subjects: **Trading and Market Microstructure (q-fin.TR)**; Mathematical Finance (q-fin.MF)

MSC classes: 60G40, 91G10, 62L15

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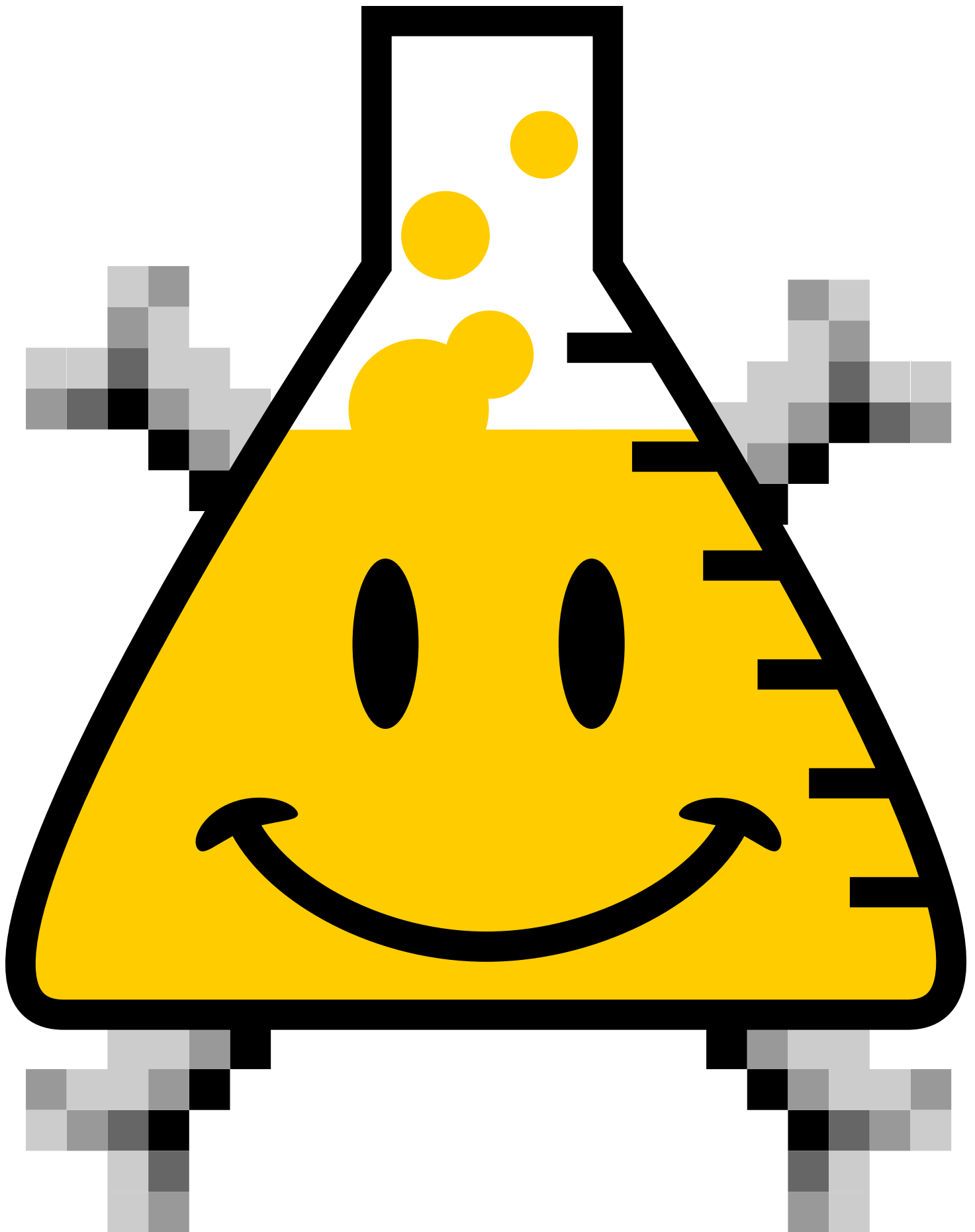
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